

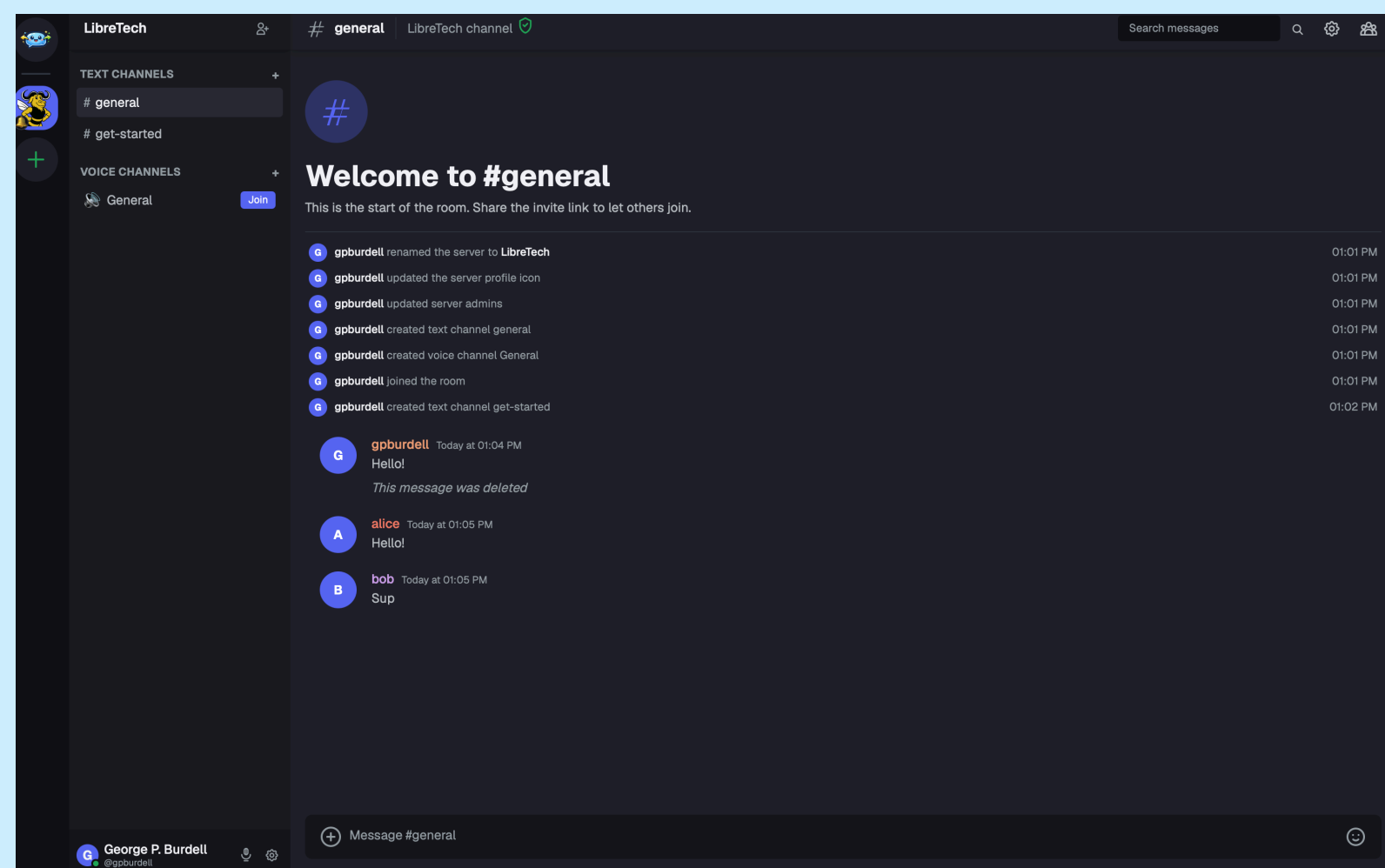


Quibble: A Peer-to-Peer Chat Application

Mentor: Dr. Jeffrey Young
Students: Isabel Aranguren | Alex Jenkins



Open Source Program Office



Project Overview

Quibble is a free and open-source peer-to-peer chat application built on the Holepunch stack. It operates without a central server— all data lives on users' machines using Hypercore.

Goal & Milestones

- * Real-time text, voice/video calls with screen share, file sharing
- * Offline delivery, a peer who's away catches up on full room history the moment they reconnect
- * Test suite covers the call lifecycle and chat message flow.

Cryptographic Identity

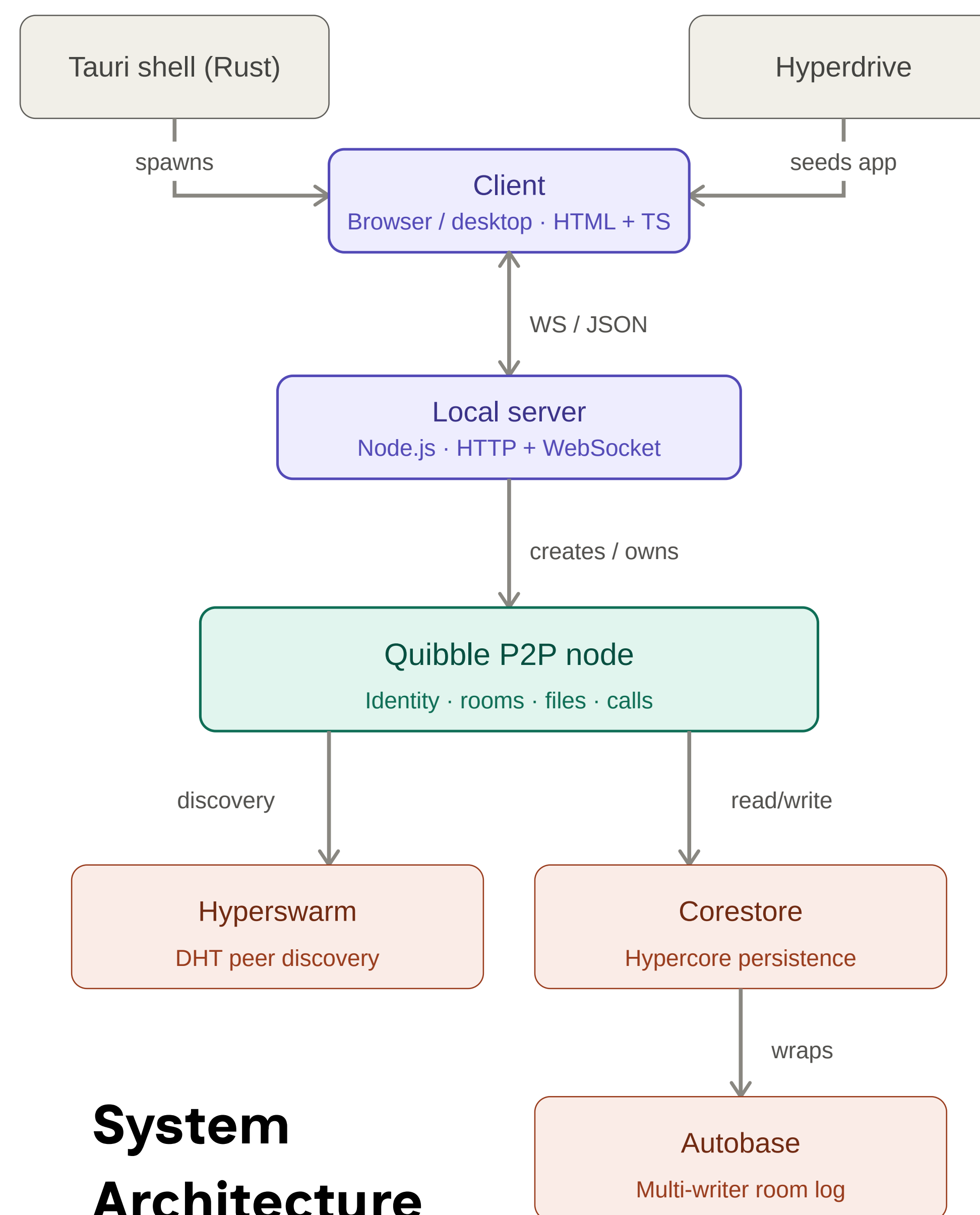
Each peer generates its own keypair on first launch. no account, no registration, no server issuing credentials.

Hyperswarm DHT

o find another peer it looks them up in a distributed hash table that every participating peer helps maintain.

Autobase

Each peer writes to its own append-only log; Autobase merges every peer's log into one consistent, causally-ordered room history.



System Architecture

Open Source Outcomes

Developer documentation covering environment setup and how to run the dual local dev clients, so a new contributor can get oriented without asking someone directly.

Architecture documentation of the full Hyperswarm + Autobase system (boot → discovery → protocol → sync), so someone can understand how the pieces fit before touching code.

Issue tracking, built out on Codeberg where none existed. Tested the app to file real, actionable issues, giving the project a working intake process for bugs and giving contributors visible entry points.

CI and review workflows on Codeberg, so contributions from people outside the original team can actually be tested and merged.

Highlights & Accomplishments

23 PRs merged into main since

Real-time voice & video calling. WebRTC mesh calling with screen share, migrated from Autobase-routed signaling to a dedicated Protomux channel for lower latency and better reliability,

Cross-platform desktop app. Packaged with Tauri (Rust) for native Linux/macOS/Windows builds.

Friends system. Cryptographically authenticated friend requests, a lifecycle state machine for friend/DM state, persisted friend profiles (avatars/display names) and hardened message handling to reject unsolicited or replayed friend-protocol messages.

Set up a docs site, so setup and architecture docs are easy to find and browse rather than buried in the repo

Chat UI & moderation, threads, message deletion, media lightbox, and per-room/per-user mute preferences.

CI/CD pipeline. Every merge to main that passes tests now automatically cross-builds a Linux Appliance and Windows installer and publishes a tagged release with auto-generated note.

Future Work

- Broader load-testing of the Autobase-based room model at higher peer counts.
- Expand moderation tooling (mute preferences and mention detection currently ship; role-based .
- Portable identities.
- Grow the contributor base beyond the current core team via the LibreTech Collective community.

Quibble aims to replace traditional chat and messaging platforms - from IRC to Discord and Slack - by building a fully free and user-owned ecosystem.

<https://docs.quibble.chat/en/latest/>
<https://codeberg.org/LTC-GT/Quibble>

